

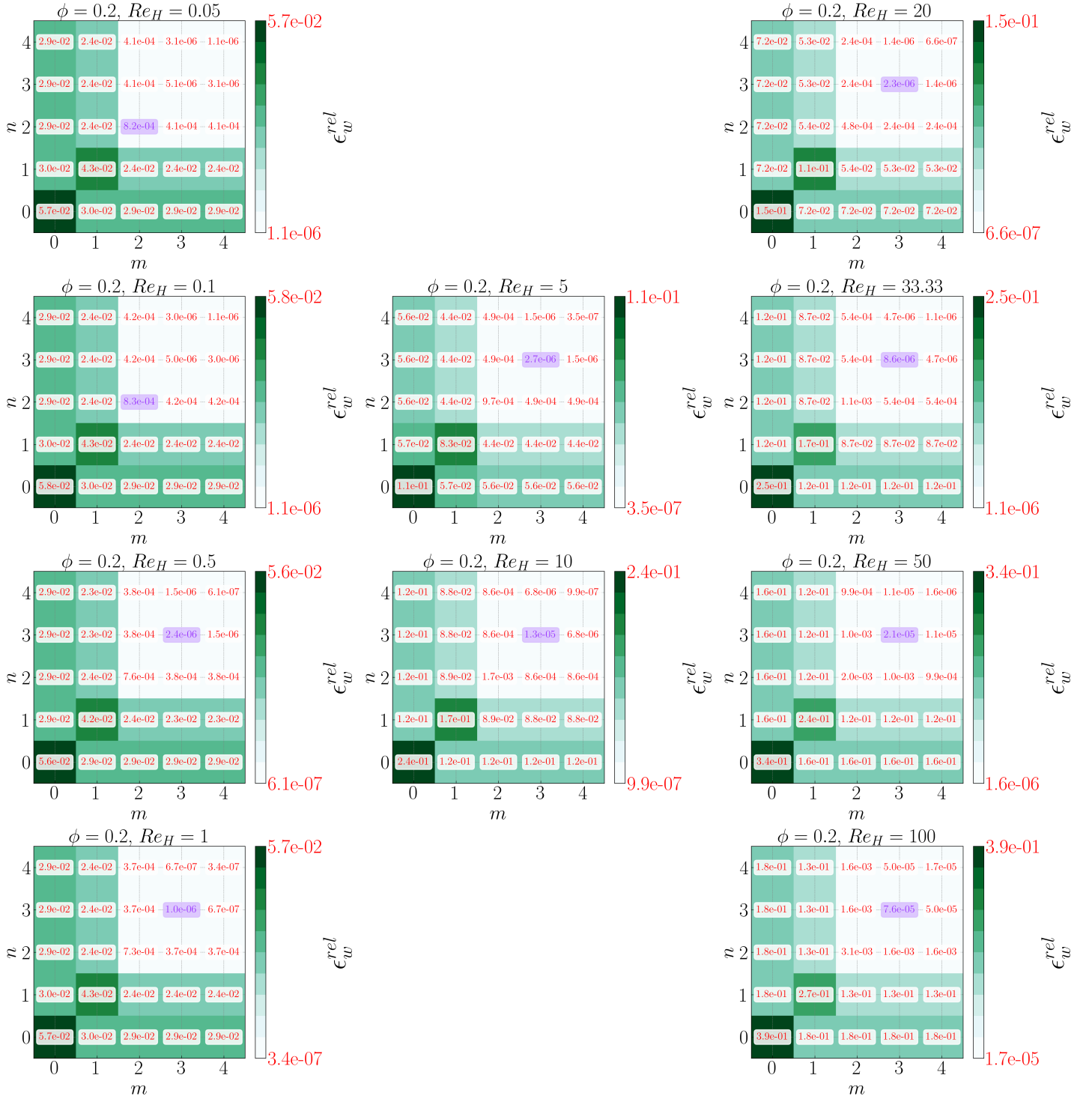
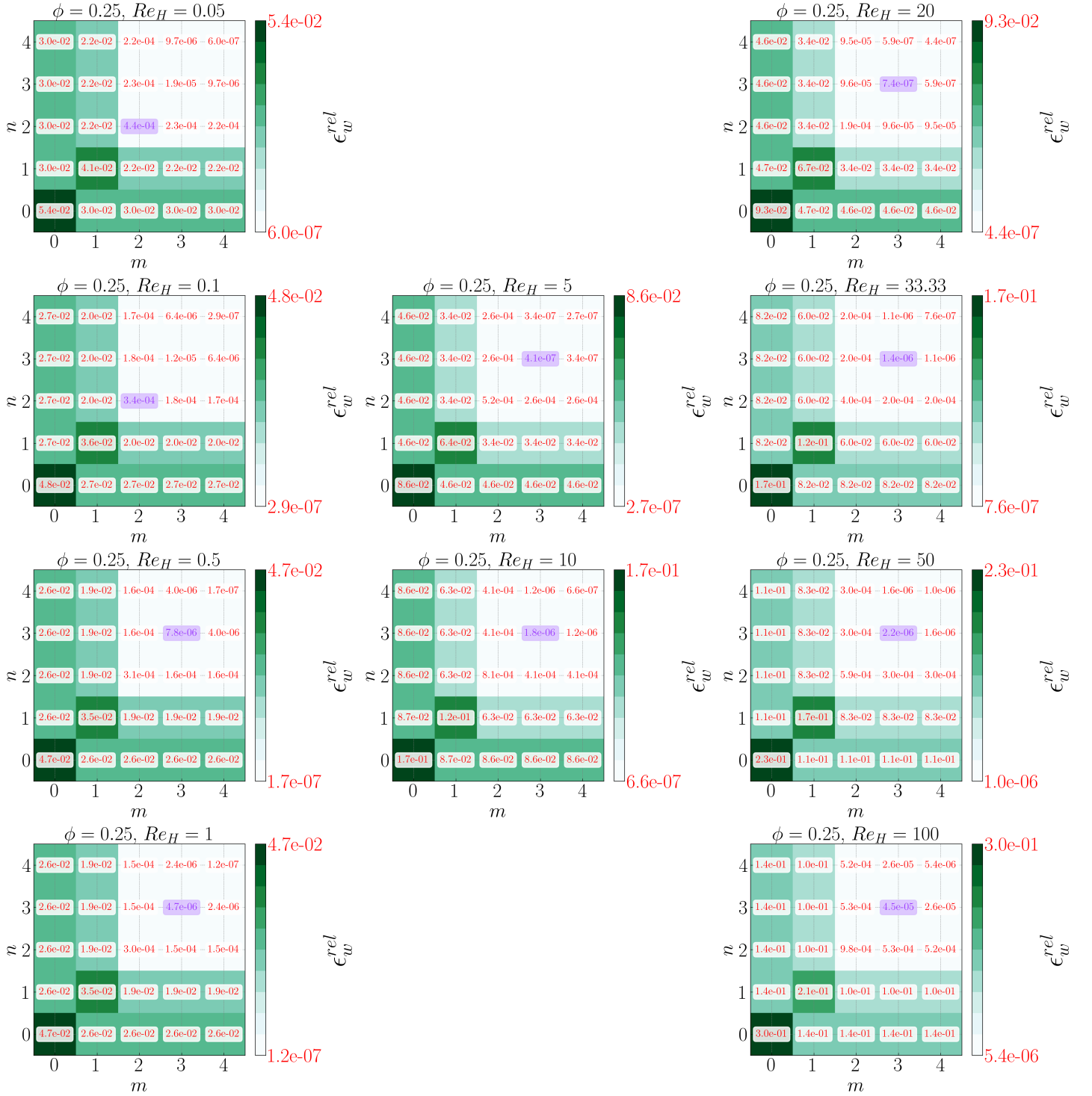


Figure B.93: Relative reconstruction error of the streamwise velocity component  $w$  obtained using Truncated Inverse Discrete Fourier Transform (TIDFT) representations for the  $1 \times 1$  SRG at  $\phi = 0.2$ . Each panel shows  $\epsilon_w^{\text{rel}}$  as a function of the retained modal truncation ( $m_T, n_T$ ) at the indicated Reynolds number. The error is evaluated within the effective downstream redistribution region,  $z \geq \Gamma/2 + \mathcal{L}_{\text{peak}}$ . Together, the panels assess the modal support required to reconstruct the corresponding downstream field over the investigated Reynolds-number range.



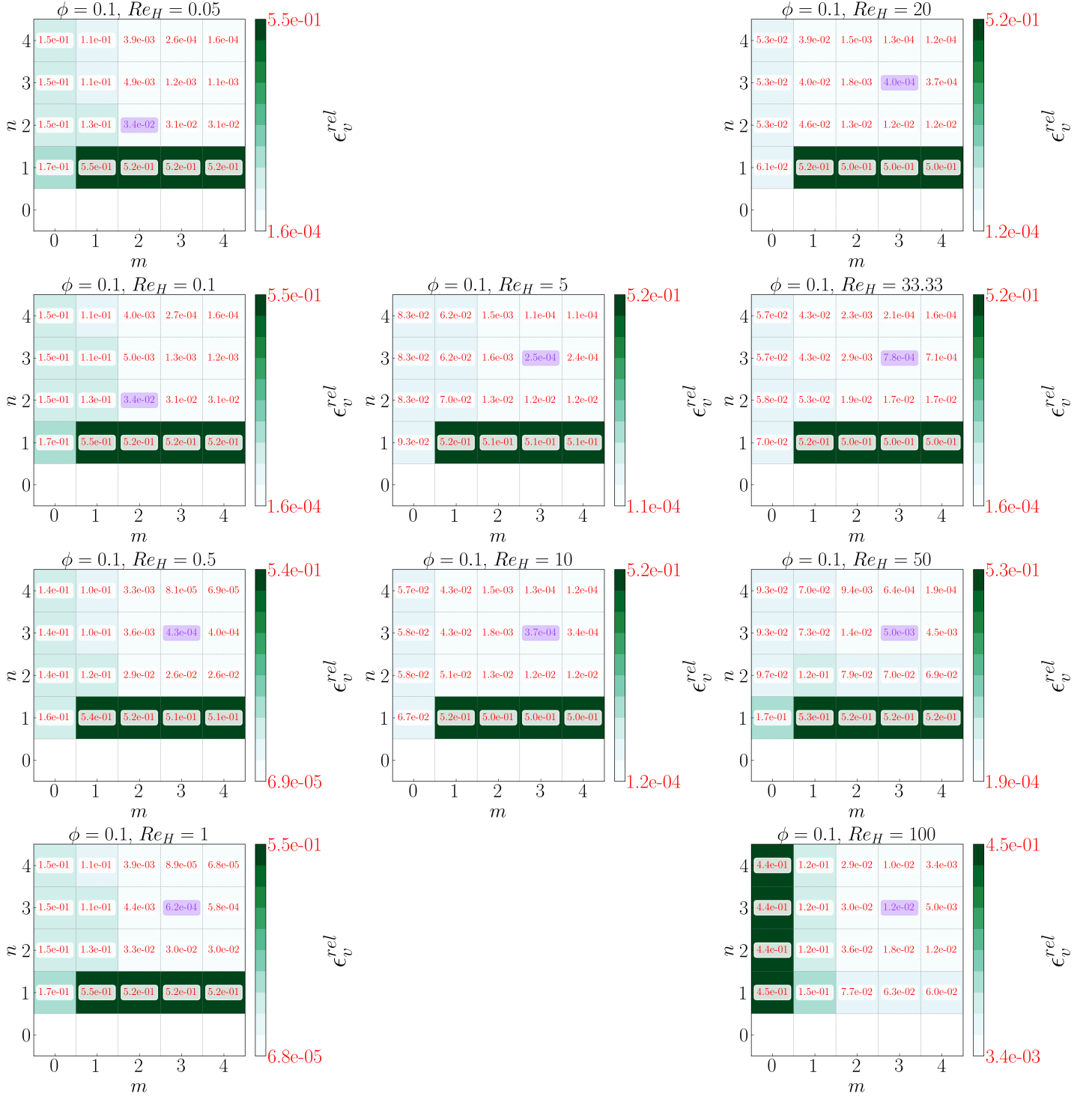


Figure B.95: Relative reconstruction error of the transverse velocity component  $v$  obtained using Truncated Inverse Discrete Fourier Transform (TIDFT) representations for the  $1 \times 1$  SRG at  $\phi = 0.1$ . Each panel shows  $\epsilon_v^{rel}$  as a function of the retained modal truncation  $(m_T, n_T)$  at the indicated Reynolds number. The error is evaluated within the effective downstream redistribution region,  $z \geq \Gamma/2 + \mathcal{L}_{\text{peak}}$ . Together, the panels assess the modal support required to reconstruct the corresponding downstream field over the investigated Reynolds-number range.

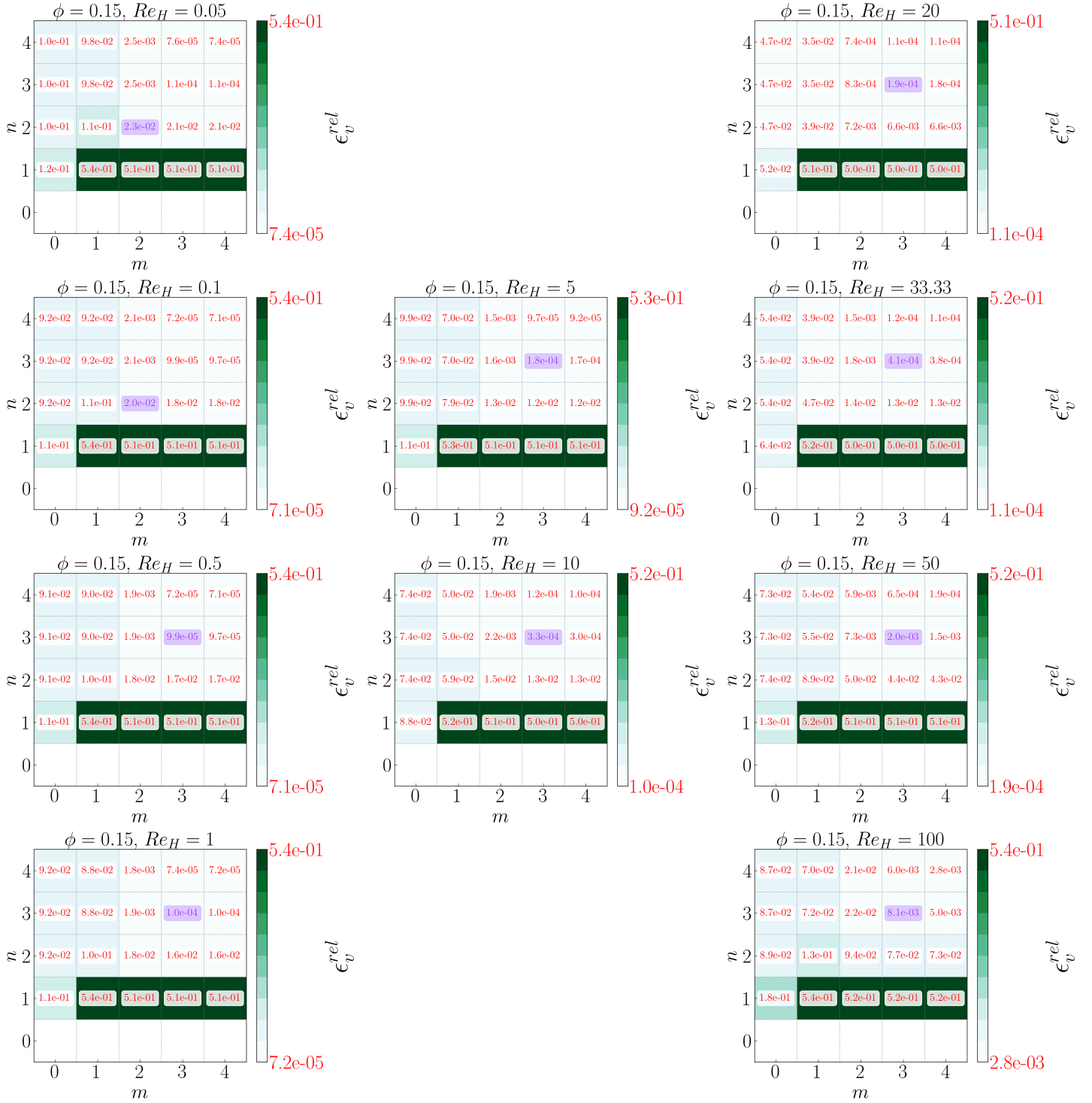


Figure B.96: Relative reconstruction error of the transverse velocity component  $v$  obtained using Truncated Inverse Discrete Fourier Transform (TIDFT) representations for the  $1 \times 1$  SRG at  $\phi = 0.15$ . Each panel shows  $\epsilon_v^{rel}$  as a function of the retained modal truncation  $(m_T, n_T)$  at the indicated Reynolds number. The error is evaluated within the effective downstream redistribution region,  $z \geq \Gamma/2 + \mathcal{L}_{\text{peak}}$ . Together, the panels assess the modal support required to reconstruct the corresponding downstream field over the investigated Reynolds-number range.



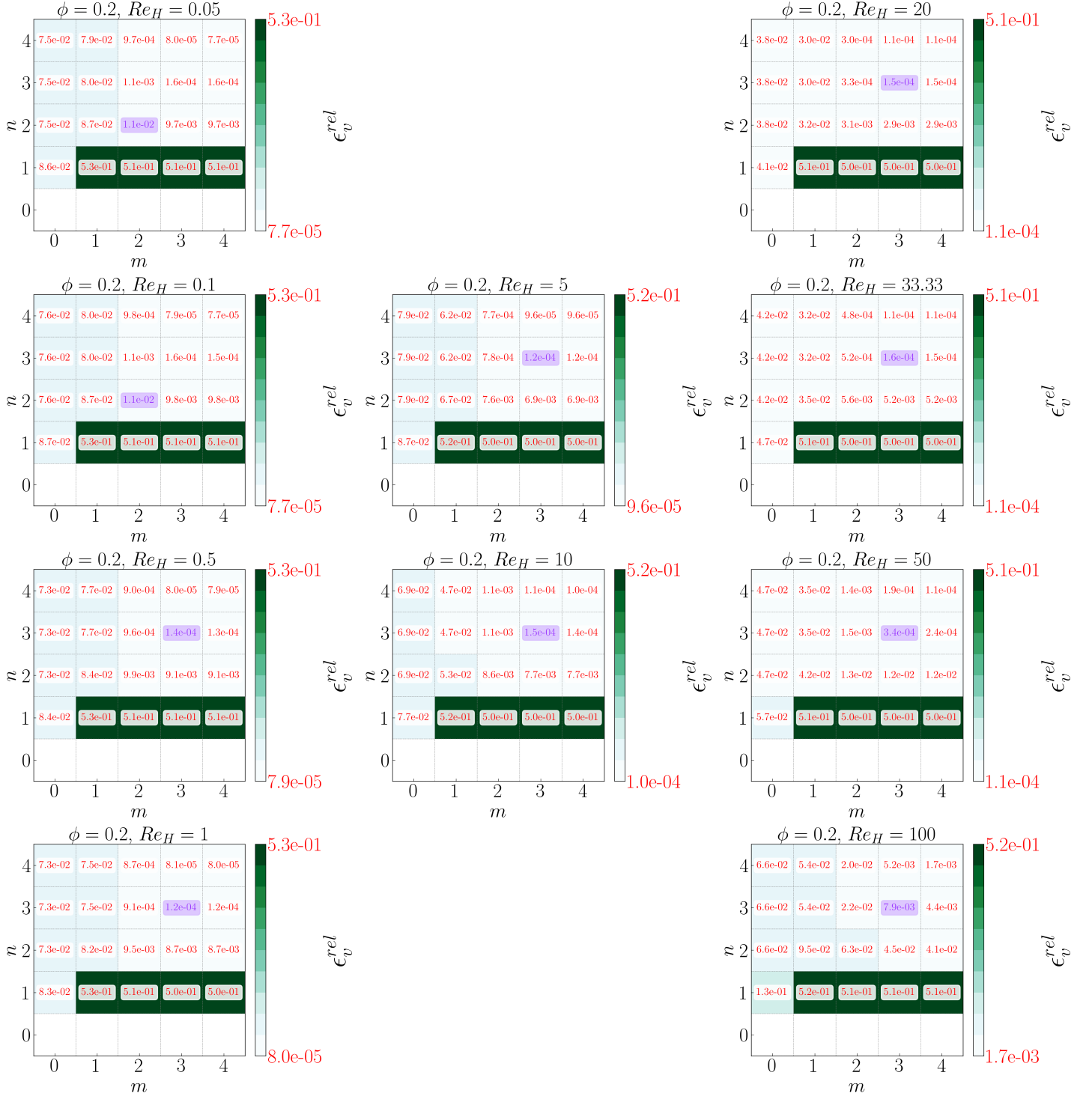


Figure B.97: Relative reconstruction error of the transverse velocity component  $v$  obtained using Truncated Inverse Discrete Fourier Transform (TIDFT) representations for the  $1 \times 1$  SRG at  $\phi = 0.2$ . Each panel shows  $\epsilon_v^{\text{rel}}$  as a function of the retained modal truncation  $(m_T, n_T)$  at the indicated Reynolds number. The error is evaluated within the effective downstream redistribution region,  $z \geq \Gamma/2 + \mathcal{L}_{\text{peak}}$ . Together, the panels assess the modal support required to reconstruct the corresponding downstream field over the investigated Reynolds-number range.

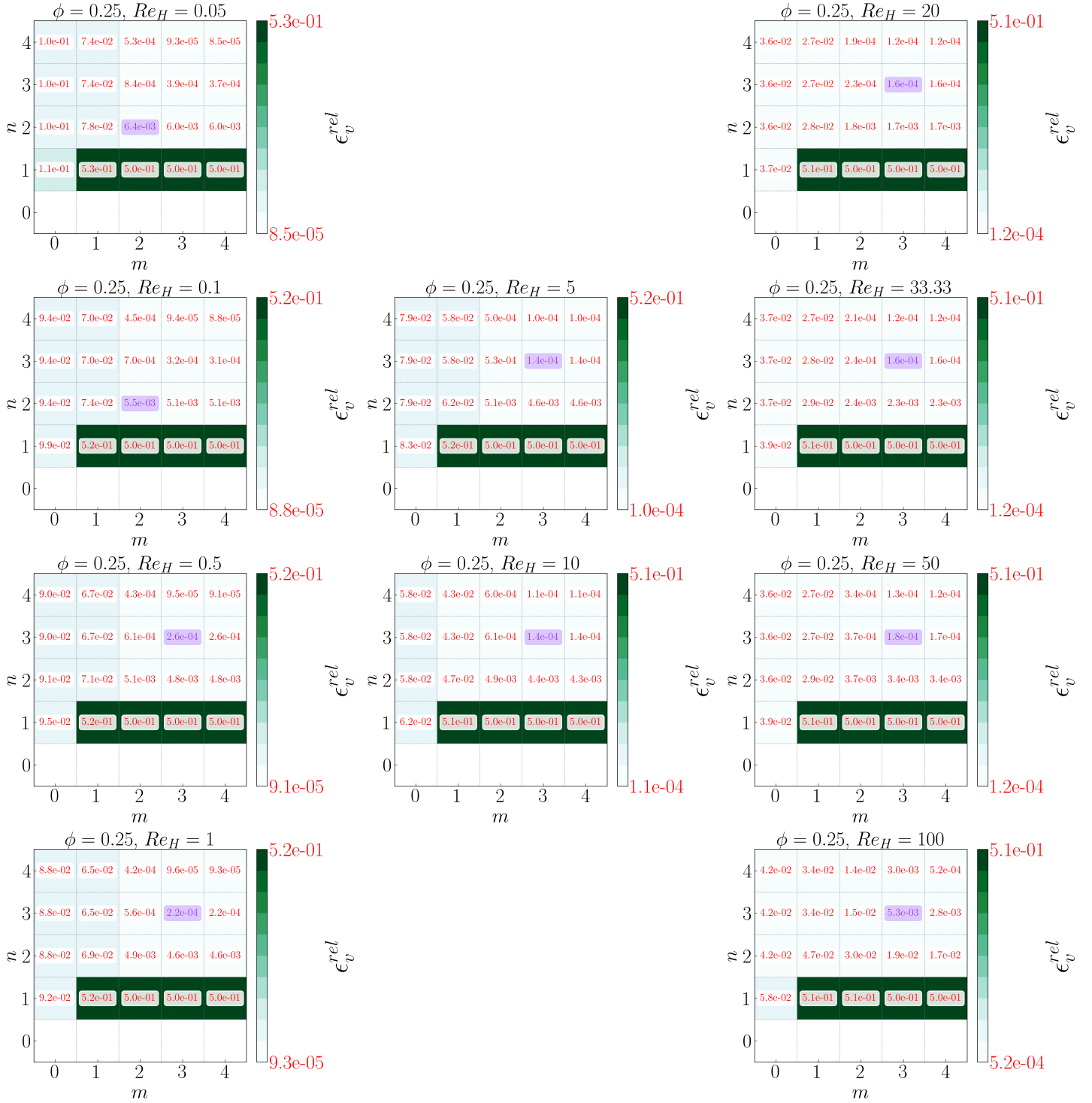


Figure B.98: Relative reconstruction error of the transverse velocity component  $v$  obtained using Truncated Inverse Discrete Fourier Transform (TIDFT) representations for the  $1 \times 1$  SRG at  $\phi = 0.25$ . Each panel shows  $\epsilon_v^{\text{rel}}$  as a function of the retained modal truncation  $(m_T, n_T)$  at the indicated Reynolds number. The error is evaluated within the effective downstream redistribution region,  $z \geq \Gamma/2 + \mathcal{L}_{\text{peak}}$ . Together, the panels assess the modal support required to reconstruct the corresponding downstream field over the investigated Reynolds-number range.

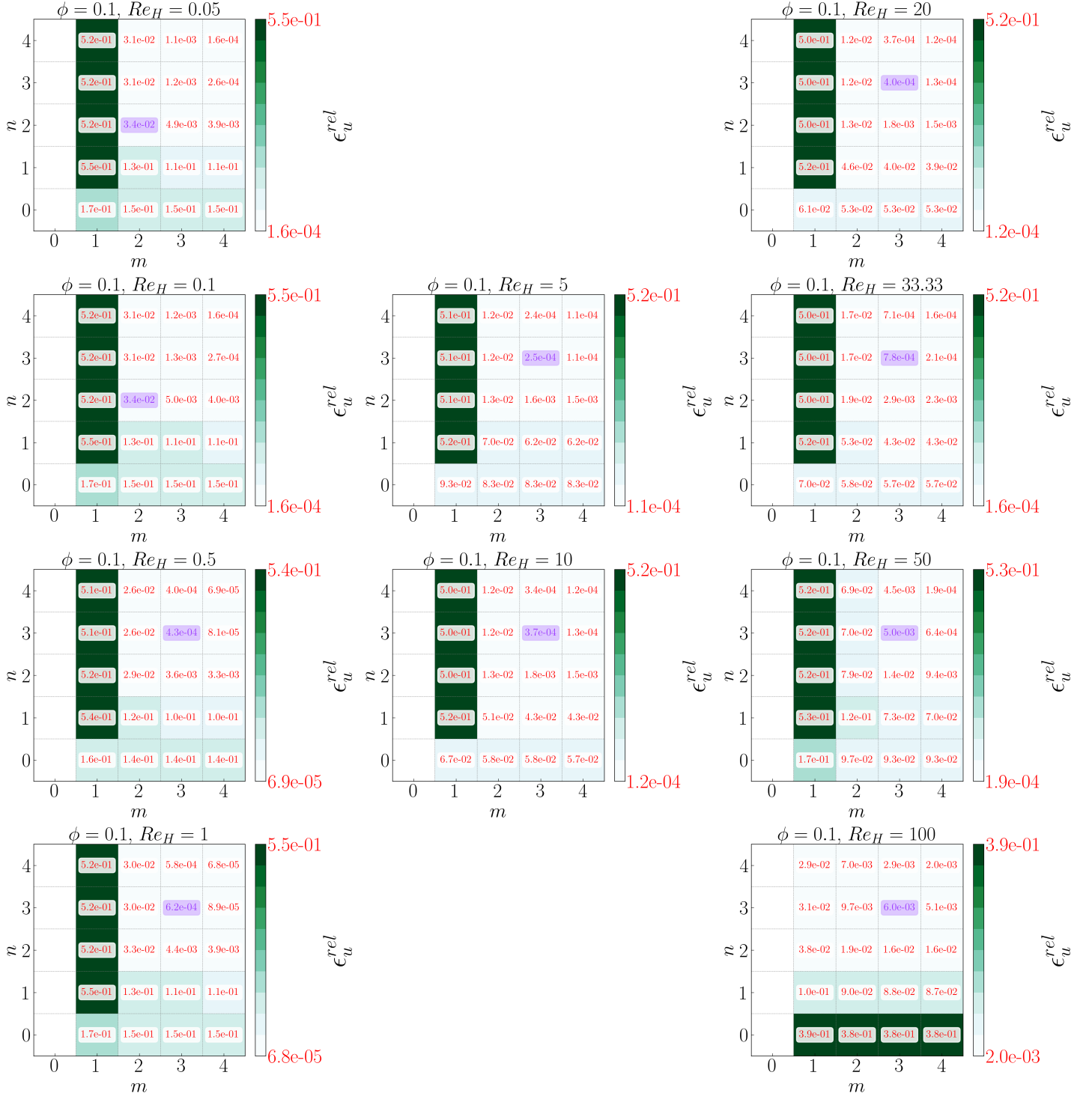


Figure B.99: Relative reconstruction error of the spanwise velocity component  $u$  obtained using Truncated Inverse Discrete Fourier Transform (TIDFT) representations for the  $1 \times 1$  SRG at  $\phi = 0.1$ . Each panel shows  $\epsilon_u^{rel}$  as a function of the retained modal truncation  $(m_T, n_T)$  at the indicated Reynolds number. The error is evaluated within the effective downstream redistribution region,  $z \geq \Gamma/2 + \mathcal{L}_{peak}$ . Together, the panels assess the modal support required to reconstruct the corresponding downstream field over the investigated Reynolds-number range.

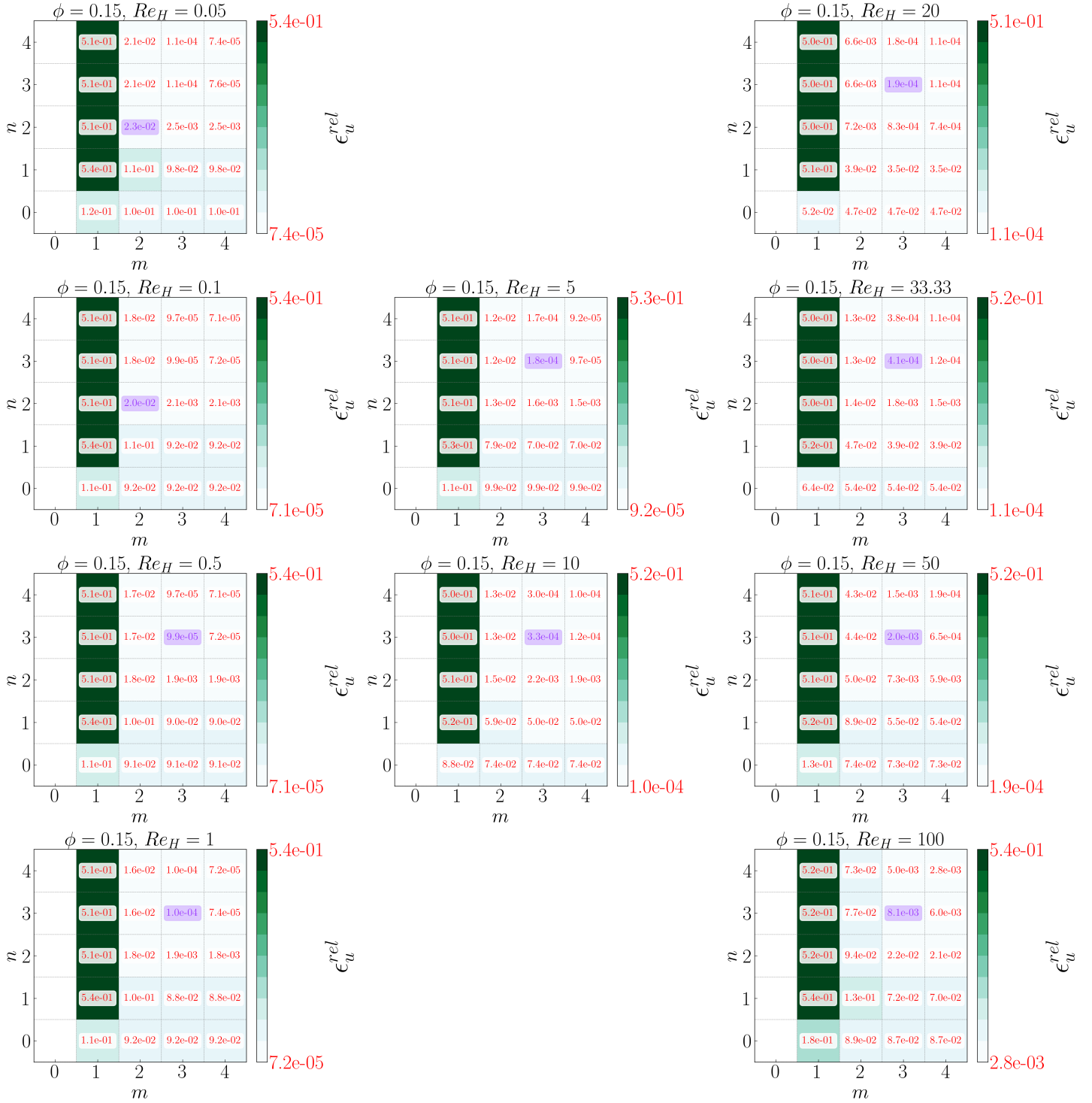


Figure B.100: Relative reconstruction error of the spanwise velocity component  $u$  obtained using Truncated Inverse Discrete Fourier Transform (TIDFT) representations for the  $1 \times 1$  SRG at  $\phi = 0.15$ . Each panel shows  $\epsilon_u^{rel}$  as a function of the retained modal truncation  $(m_T, n_T)$  at the indicated Reynolds number. The error is evaluated within the effective downstream redistribution region,  $z \geq \Gamma/2 + \mathcal{L}_{\text{peak}}$ . Together, the panels assess the modal support required to reconstruct the corresponding downstream field over the investigated Reynolds-number range.

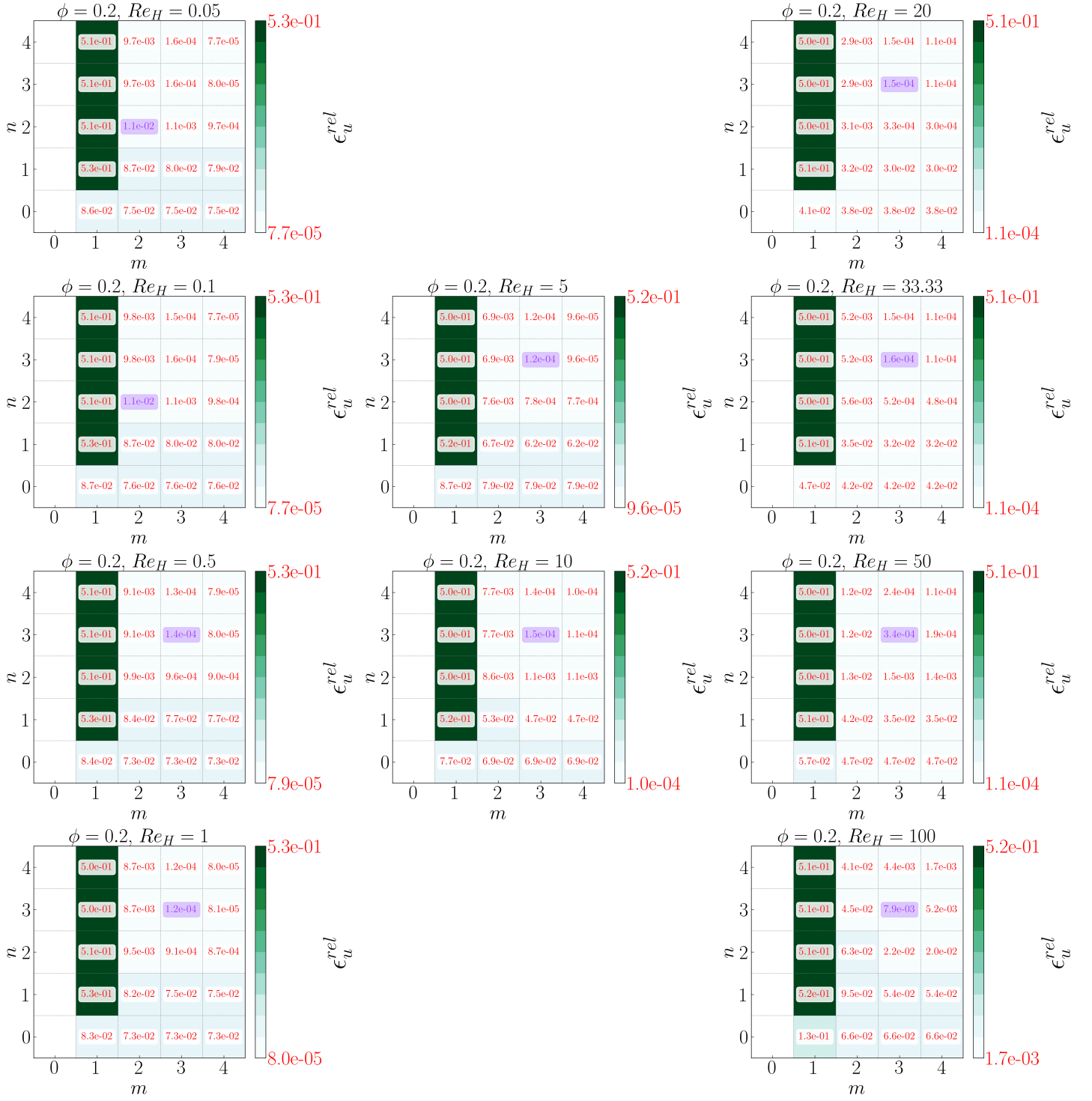


Figure B.101: Relative reconstruction error of the spanwise velocity component  $u$  obtained using Truncated Inverse Discrete Fourier Transform (TIDFT) representations for the  $1 \times 1$  SRG at  $\phi = 0.2$ . Each panel shows  $\epsilon_u^{\text{rel}}$  as a function of the retained modal truncation  $(m_T, n_T)$  at the indicated Reynolds number. The error is evaluated within the effective downstream redistribution region,  $z \geq \Gamma/2 + \mathcal{L}_{\text{peak}}$ . Together, the panels assess the modal support required to reconstruct the corresponding downstream field over the investigated Reynolds-number range.

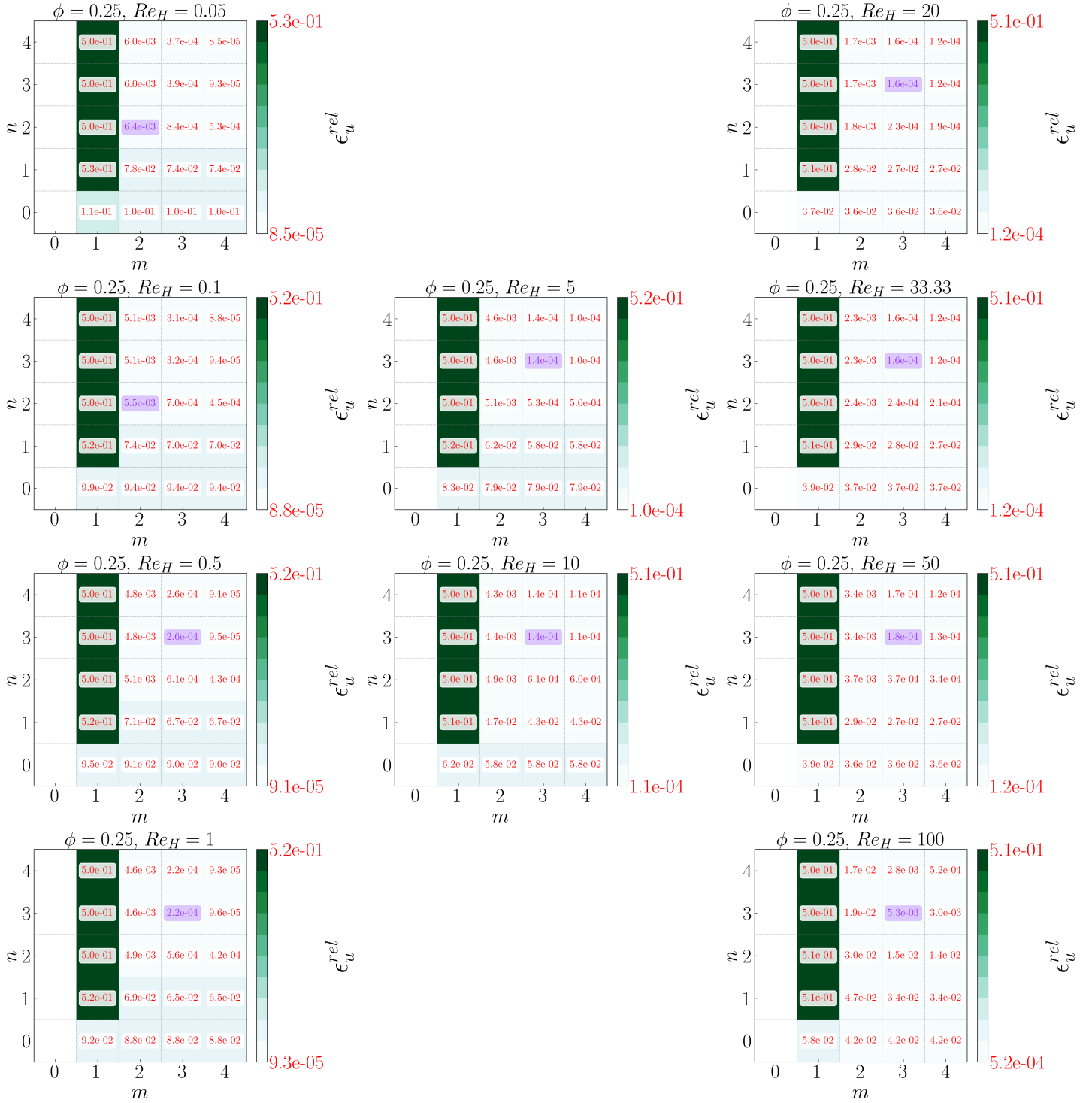


Figure B.102: Relative reconstruction error of the spanwise velocity component  $u$  obtained using Truncated Inverse Discrete Fourier Transform (TIDFT) representations for the  $1 \times 1$  SRG at  $\phi = 0.25$ . Each panel shows  $\epsilon_u^{\text{rel}}$  as a function of the retained modal truncation  $(m_T, n_T)$  at the indicated Reynolds number. The error is evaluated within the effective downstream redistribution region,  $z \geq \Gamma/2 + \mathcal{L}_{\text{peak}}$ . Together, the panels assess the modal support required to reconstruct the corresponding downstream field over the investigated Reynolds-number range.



Figure B.103: Relative reconstruction error of the streamwise force component  $F_z$  obtained using Truncated Inverse Discrete Fourier Transform (TIDFT) representations for the  $1 \times 1$  SRG at  $\phi = 0.1$ . Each panel shows  $\epsilon_{F_z}^{\text{rel}}$  as a function of the retained modal truncation  $(m_T, n_T)$  at the indicated Reynolds number. The error is evaluated within the effective downstream redistribution region,  $z \geq \Gamma/2 + \mathcal{L}_{\text{peak}}$ . Together, the panels assess the modal support required to reconstruct the corresponding downstream field over the investigated Reynolds-number range.

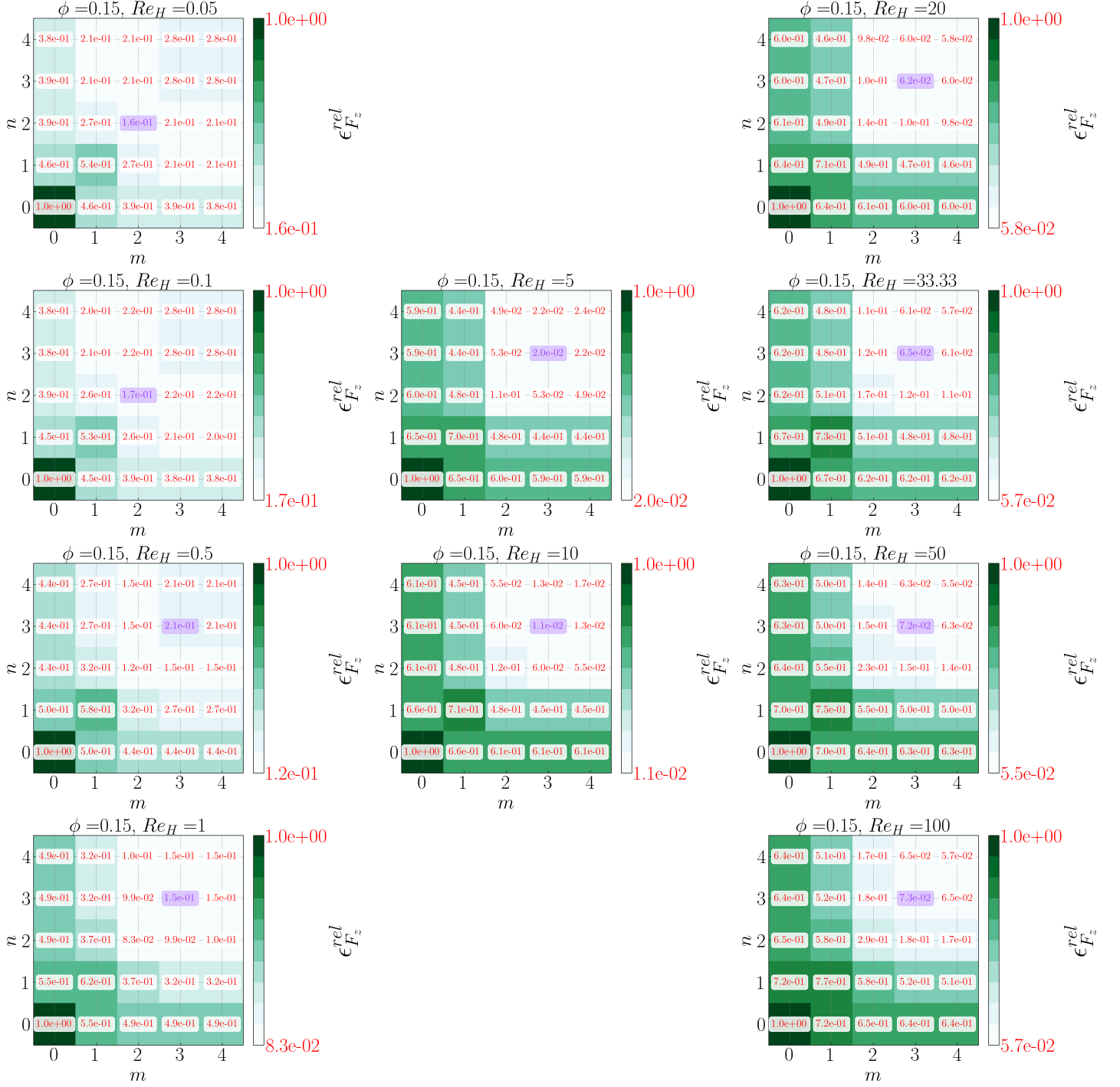


Figure B.104: Relative reconstruction error of the streamwise force component  $F_z$  obtained using Truncated Inverse Discrete Fourier Transform (TIDFT) representations for the  $1 \times 1$  SRG at  $\phi = 0.15$ . Each panel shows  $\epsilon_{F_z}^{rel}$  as a function of the retained modal truncation  $(m_T, n_T)$  at the indicated Reynolds number. The error is evaluated within the effective downstream redistribution region,  $z \geq \Gamma/2 + \mathcal{L}_{\text{peak}}$ . Together, the panels assess the modal support required to reconstruct the corresponding downstream field over the investigated Reynolds-number range.





Figure B.105: Relative reconstruction error of the streamwise force component  $F_z$  obtained using Truncated Inverse Discrete Fourier Transform (TIDFT) representations for the  $1 \times 1$  SRG at  $\phi = 0.2$ . Each panel shows  $\epsilon_{F_z}^{\text{rel}}$  as a function of the retained modal truncation  $(m_T, n_T)$  at the indicated Reynolds number. The error is evaluated within the effective downstream redistribution region,  $z \geq \Gamma/2 + \mathcal{L}_{\text{peak}}$ . Together, the panels assess the modal support required to reconstruct the corresponding downstream field over the investigated Reynolds-number range.



Figure B.106: Relative reconstruction error of the streamwise force component  $F_z$  obtained using Truncated Inverse Discrete Fourier Transform (TIDFT) representations for the  $1 \times 1$  SRG at  $\phi = 0.25$ . Each panel shows  $\epsilon_{F_z}^{rel}$  as a function of the retained modal truncation  $(m_T, n_T)$  at the indicated Reynolds number. The error is evaluated within the effective downstream redistribution region,  $z \geq \Gamma/2 + \mathcal{L}_{\text{peak}}$ . Together, the panels assess the modal support required to reconstruct the corresponding downstream field over the investigated Reynolds-number range.

# Appendix C

## Detailed Machine-Learning Validation of Fourier Coefficients

This appendix compares Fourier coefficients extracted from the geometry-resolved small-scale-flow (SSF) simulations with their machine-learning (ML) predictions. The analysis concerns the square regular grid (SRG). The complex coefficients  $\hat{W}_{m,n}$  and  $\hat{V}_{m,n}$  represent the streamwise velocity  $w$  and transverse velocity  $v$ , respectively, on periodic planes downstream of the grid.

For the steady SRG flows used to construct the training database, the geometrical symmetries give

$$\hat{W}_{m,n} = \hat{W}_{n,m}, \quad \hat{V}_{m,n} = \hat{U}_{n,m}, \quad (\text{C.1})$$

where  $\hat{U}_{m,n}$  denotes the coefficient of the other transverse velocity component  $u$ . These relations reduce the number of independent coefficients that must be predicted.

The retained independent streamwise modes are

$$\mathcal{M}_{\hat{W}} = \left\{ \begin{array}{cccc} (0, 3), & (1, 3), & (2, 3), & (3, 3), \\ (0, 2), & (1, 2), & (2, 2), & \\ (0, 1), & (1, 1), & & \\ (0, 0) & & & \end{array} \right\}, \quad (\text{C.2})$$

and the retained independent transverse modes are

$$\mathcal{M}_{\hat{V}} = \left\{ \begin{array}{cccc} (0, 3), & (1, 3), & (2, 3), & (3, 3), \\ (0, 2), & (1, 2), & (2, 2), & (3, 2), \\ (0, 1), & (1, 1), & (2, 1), & (3, 1) \end{array} \right\}. \quad (\text{C.3})$$

The coefficients not predicted independently are recovered using the symmetry relations in Eq. (C.1) and the corresponding Fourier conjugate relations.

A separate feed-forward neural-network regressor is trained for each independent coefficient. Its inputs are the downstream coordinate  $z$ , grid porosity  $\phi$ , and Reynolds number  $\text{Re}_H$ :

$$(z, \phi, \text{Re}_H) \mapsto \hat{c}_{m,n}, \quad \hat{c} \in \{\hat{W}, \hat{V}\}. \quad (\text{C.4})$$

For each complex coefficient, the network predicts its real and imaginary parts. The zeroth streamwise coefficient  $\Re(\hat{W}_{0,0}) = w_{\text{face}}$  is real-valued and is predicted by a separate scalar-output network.

All mode-wise regressors use three hidden layers with 128 neurons per layer, GELU activations, and a linear output layer. Inputs and targets are scaled component-wise using Min–Max scaling, with the scaling parameters determined from the training subset. The models are trained for 300 epochs using the Adam optimizer, a learning rate of  $10^{-3}$ , a batch size of 256, and a mean-squared-error loss. For each regressor, 90% of the available samples are used for training and 10% are reserved for validation.

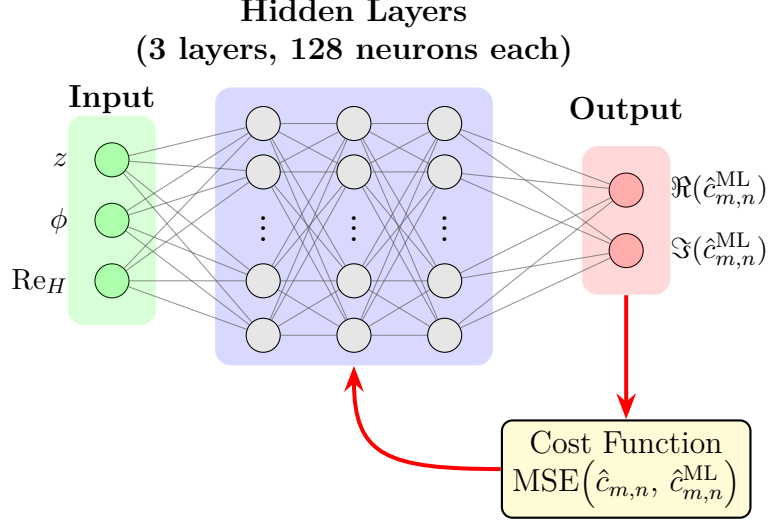


Figure C.1: Schematic of the two-output, mode-wise neural-network regressor for a complex Fourier coefficient. The inputs are  $(z, \phi, \text{Re}_H)$ ; the outputs are the real and imaginary parts of  $\hat{c}_{m,n}^{\text{ML}}$ . A separate network is trained for each independent coefficient. The remaining coefficients are recovered using the symmetry relations in Eq. (C.1). The coefficient  $\Re(\hat{W}_{0,0})$  uses a scalar-output variant.

The network parameters are optimized by minimizing the mean-squared error

$$\mathcal{J} = \frac{1}{N} \sum_{k=1}^N \|\mathbf{y}_k^{\text{ML}} - \mathbf{y}_k\|_2^2, \quad (\text{C.5})$$

where  $N$  is the number of training samples. For a complex mode,  $\mathbf{y}_k$  contains the real and imaginary parts of the reference coefficient; for  $\hat{W}_{0,0}$ , it is the scalar  $\Re(\hat{W}_{0,0})$ . The loss on the validation subset is monitored during training but is not minimized by the optimizer.

The training database consists of steady SRG configurations with

$$\begin{aligned} \phi &\in \{0.10, 0.15, 0.20, 0.25, 0.30, 0.442649\}, \\ \text{Re}_H &\in \{0.05, 0.10, 0.50, 1, 5, 10, 20, 33.33, 50\}. \end{aligned}$$

The comparisons at these parameter values assess predictions within the training parameter range. The validation subset contains samples withheld from optimization within this range; it does not represent a test at unseen porosity or Reynolds-number values.

Extrapolation is assessed separately at  $\text{Re}_H = 100$ , which is excluded from both training and validation. At  $(\phi, \text{Re}_H) = (0.10, 100)$ , the reference flow is weakly unsteady and may break the geometrical symmetries represented by Eq. (C.1). The

symmetry-constrained ML predictions for this case must therefore be interpreted with that limitation in mind.

The figures below compare the real and imaginary parts of the reference DFT coefficients with the corresponding ML predictions. They are organized by predictions within the training parameter range, extrapolation at  $Re_H = 100$ , and training histories.

## C.1 Prediction within the training parameter range

This section compares the reference DFT coefficients with ML predictions at the porosity and Reynolds-number values included in the training database. The figures report the retained independent  $\hat{W}$  and  $\hat{V}$  coefficients available in the detailed comparison set.

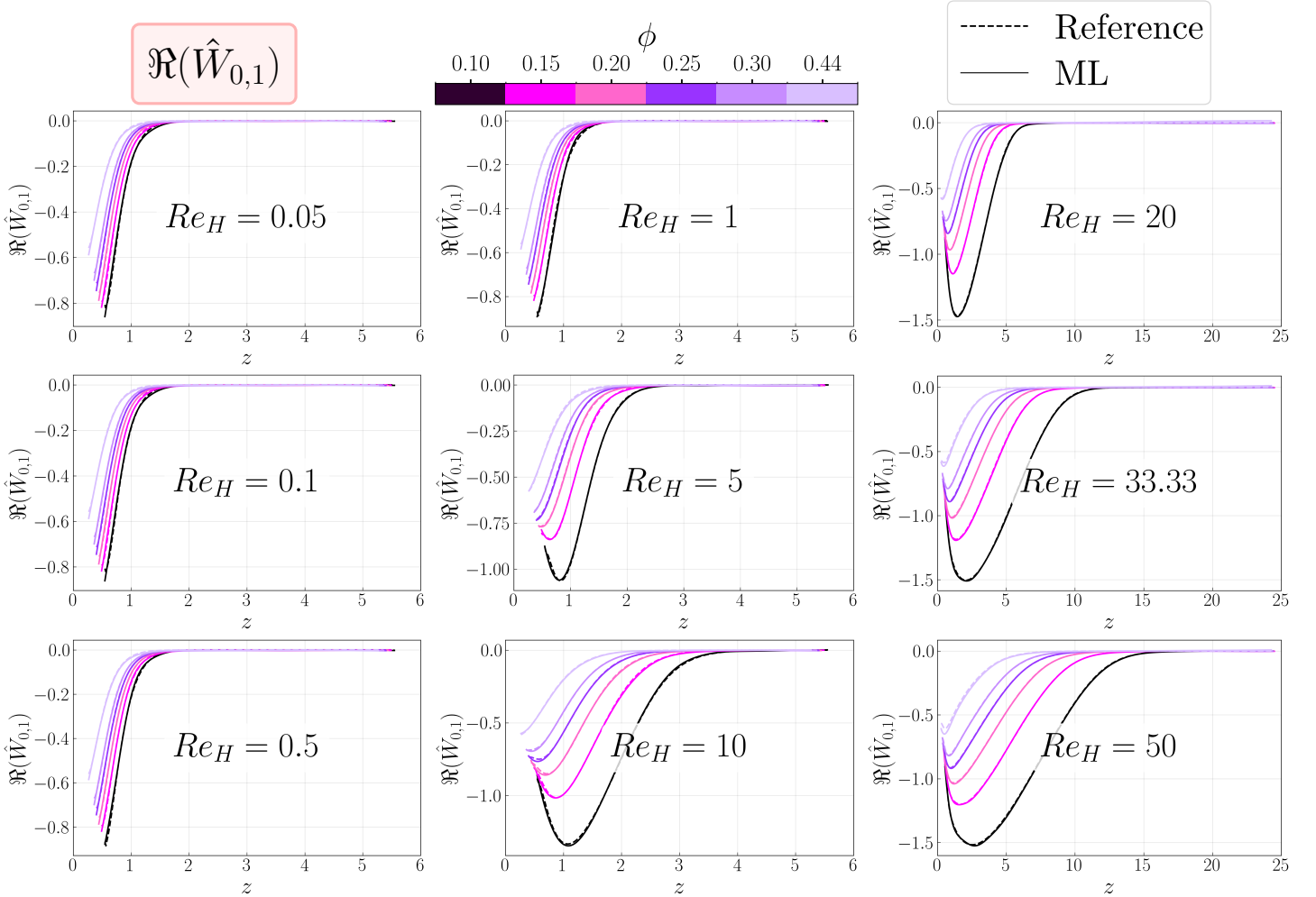


Figure C.2: Comparison of the reference DFT coefficient  $\Re(\hat{W}_{0,1})$  (dashed lines) and its neural-network prediction  $\Re(\hat{W}_{0,1}^{ML})$  (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of  $z$  at the indicated Reynolds number; colors denote the porosity  $\phi$ . All cases lie within the parameter range covered by the training database.